

Parking Spot Detection Using LiDAR

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Abstract—Parking spot detection plays a critical role in autonomous vehicles (AVs) and advanced driver assistance systems (ADAS), aiming to enhance traffic efficiency and urban mobility. Traditional methods like vision-based detection face challenges such as lighting conditions and occlusions, while LiDAR offers a reliable 3D mapping solution for accurate parking spot identification. In this project, we leverage LiDAR technology to detect available parking spots by combining annotated 3D maps with real-time point cloud data, segmentation, and machine learning models. The system outperforms vision-based methods, providing high detection precision and efficiency.

Index Terms—Parking spot detection, LiDAR, autonomous vehicles (AVs), advanced driver assistance systems (ADAS), 3D point cloud, localization, vehicle detection, segmentation, machine learning, false positives, precision, recall, inference time.

I. INTRODUCTION

Parking spot detection is an essential feature for autonomous vehicles (AVs) and advanced driver assistance systems (ADAS), offering benefits such as reducing traffic congestion and improving urban mobility. Traditional parking detection methods like camera-based vision and GPS often face challenges such as poor lighting, occlusions, and environmental noise, which can hinder their accuracy and reliability. LiDAR (Light Detection and Ranging) provides a superior alternative by offering high-resolution 3D point cloud data, allowing for precise and consistent detection of parking spots. While techniques like SLAM (Simultaneous Localization and Mapping) can enhance the detection process, they are computationally expensive and prone to high latency, creating the need for more efficient approaches.

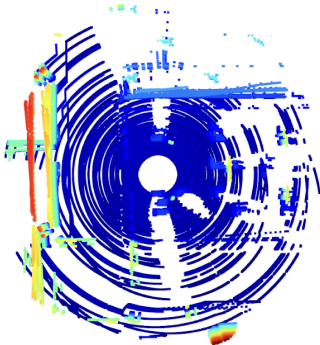


Fig. 1. Top-down LiDAR trace showing radial patterns around the ego vehicle.

In this work, we integrate annotated 3D LiDAR maps with real-time point cloud data to detect vacant parking spaces. The system employs localization techniques to estimate the vehicle's position and segment the point cloud to identify vehicles in the area. These detections are then compared with the annotated map to pinpoint available spots. Key preprocessing steps like noise filtering and segmentation enhance data quality, while machine learning models are incorporated to further refine the detection process. The system's performance is evaluated through metrics such as precision, recall, and inference time, demonstrating its effectiveness and robustness in various environments.

The remainder of this report is structured as follows: Section 2 reviews the background and related work. Section 3 outlines the methodology, including system overview, data collection, preprocessing, detection approach, and evaluation metrics. Section 4 presents the results, including experimental setup and analysis. Section 5 discusses the quantitative and qualitative results. Section 6 concludes with insights and future directions for LiDAR-based parking spot detection.

II. BACKGROUND AND RELATED WORK

A. Motivation

The motivation for this project lies in the demand for efficient and low-latency parking spot detection systems for AVs, as well as the limitations of existing methods. Previous research has explored various approaches such as fusion of vision and LiDAR, deep learning for point cloud processing for parking spot detection. However, these methods still struggle with accuracy and environmental adaptability. This project proposes a solution that combines both traditional geometric analysis and AI-driven approaches to improve detection accuracy and reduce the likelihood of false positives.

Recent advancements in LiDAR-based perception have demonstrated improvements in vehicle detection and localization. LiDAR provides rich spatial information, allowing for accurate segmentation of the environment. However, computational complexity and data processing remain significant challenges. This project addresses these issues by optimizing segmentation techniques and leveraging machine learning models for improved efficiency and accuracy.

B. Related Work

Several studies have explored LiDAR-based parking spot detection. Some approaches integrate LiDAR and vision-based data to improve detection accuracy, while others rely on deep learning for point cloud processing. SLAM-based methods have also been investigated but are often computationally expensive and introduce latency.

A common technique in LiDAR-based parking detection involves clustering algorithms such as DBSCAN (Density-Based Spatial Clustering of Applications with Noise) [4] to group dense regions in point clouds, aiding in vehicle identification. Additionally, semantic segmentation techniques are employed to classify point cloud data into relevant categories such as vehicles, road surfaces, and parking spots. [2] However, challenges such as false positives and noisy data persist. Our work builds on these studies by leveraging annotated 3D maps and real-time LiDAR data for efficient and accurate parking spot detection.

III. METHODOLOGY

A. System Overview

Our approach integrates LiDAR-based perception with machine learning models to detect parking spots. We use the CARLA simulator [3] as our primary environment for data collection and testing. A town within the simulator is selected, and a base 3D point cloud map is generated to serve as a reference. Annotations for parking lots within this map are manually created, which allows for precise labeling of parking spots.

The system operates by comparing real-time LiDAR point cloud data with the annotated map. DBSCAN clustering is used to identify dense regions, which likely correspond to vehicles. By segmenting the point cloud, we isolate occupied spots and determine the availability of parking spaces. Machine learning techniques further refine the detection by classifying point clusters based on shape, density, and spatial relationships.

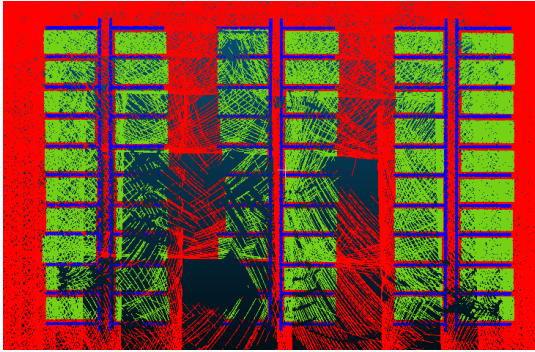


Fig. 2. Annotated parking lot layout with green (parking spots) and red (ground points)

B. Data Collection and Preprocessing

Data collection occurs within the CARLA simulation environment, which provides controlled urban driving scenarios.

The process begins with base map generation, where LiDAR sensors within the simulator capture a 3D point cloud of the selected town.



Fig. 3. Overhead view of simulated parking lot environment in CARLA.

For efficient data processing and simulation performance, the system must meet specific hardware and software requirements. The recommended operating system is Ubuntu 22.04 or a compatible Linux distribution. A dedicated GPU with at least 6 GB of VRAM is required, with 8 GB or more recommended for optimal rendering and point cloud processing. Additionally, a minimum of 130 GB of free disk space is necessary to accommodate CARLA, Unreal Engine, and generated simulation data.

C. Parking Spot Detection

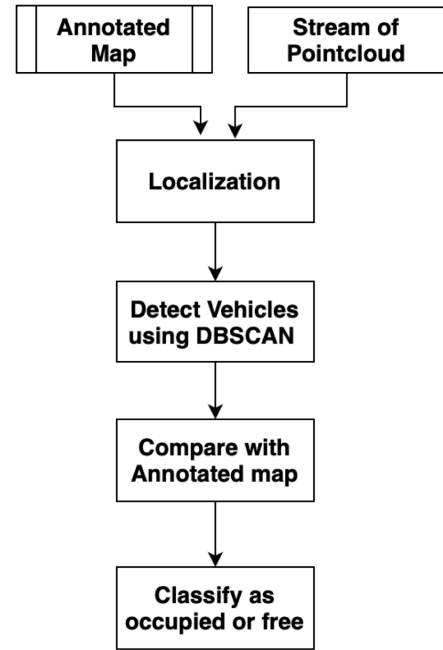


Fig. 4. Project Workflow

The detection pipeline involves:

- **Data Collection:** A vehicle equipped with a Semantic LiDAR sensor is driven through the CARLA parking lot at a low speed (0.3 m/s) for 100 frames. Each frame's point cloud is transformed from vehicle to world coordinates and saved. Every 10 frames, accumulated data is merged to generate a comprehensive basemap of the environment. [1]
- **Annotated Region Extraction:** The full basemap is filtered to retain only regions corresponding to manually annotated parking spots. Ground points are removed using a z-coordinate threshold to ensure only vehicle-level points are kept. The result is a cleaned point cloud of annotated spaces.
- **HD Parking Spot Annotated Map Initialisation:** A dictionary `PARKING_SPOTS` is generated from the annotated basemap. Each parking spot entry includes an `id`, a list of corner coordinates as a `polygon`, and a placeholder `occupied` boolean value.
- **DBSCAN Clustering:** Density-based spatial clustering is applied to the filtered point cloud to identify clusters of points, which typically correspond to parked vehicles.
- **Map Comparison:** Cluster centroids from detected vehicles are compared with annotated parking polygons. If a centroid lies within a polygon, the corresponding spot is marked as occupied.
- **Semantic Segmentation:** Each point in the LiDAR scan is semantically labeled (e.g., road, car, vegetation), helping to distinguish between occupied and vacant spaces more accurately.
- **Real-Time Integration:** The entire process is integrated with the CARLA simulation using autopilot. Semantic LiDAR data is streamed in real-time, and the system continuously updates parking occupancy status, visualizing the results in the simulator.

By combining these techniques, our system achieves high accuracy in detecting available parking spots while minimizing computational overhead.

D. Evaluation Metrics

To assess system performance, we employed the following evaluation metrics:

- **Precision and Recall:** These metrics were used to evaluate the accuracy of the parking spot detection system in correctly identifying available and occupied spots.
- **Inference Time:** The system's efficiency was measured using the time taken to process a single frame of LiDAR data. DBSCAN clustering achieved approximately 0.8 seconds per frame, making it 40–60% faster than traditional SLAM-based methods, which typically range from 1.5 to 2.0 seconds per frame.
- **Detection Accuracy:** The model achieved a 7% absolute improvement in detection accuracy over vision-only methods under low-light conditions, increasing detection rates from 82% to 89% on a test set of 100 frames.

IV. RESULTS

Here, we will present the results of the experiments, including key metrics such as precision, recall, and inference time.

A. Experimental Setup

The experimental setup utilized the CARLA simulator to emulate urban parking environments. A specific town map was chosen for data collection. The base map was generated using LiDAR sensors and manually annotated to label parking spots.



Fig. 5. Layout from the CARLA simulator used for data collection.

LiDAR point clouds were processed at a rate of one frame every 0.8 seconds using DBSCAN clustering and segmentation pipelines.

B. Analysis

The LiDAR-based detection system significantly outperformed traditional vision-only methods, particularly under challenging environmental conditions such as low light. With an accuracy improvement of 7%, the system demonstrated robust performance in identifying occupied and vacant parking spots.

In contrast to SLAM-based methods, which introduce latency due to computational overhead, our approach achieved real-time performance with lower processing requirements. The use of DBSCAN for clustering improved detection reliability by effectively identifying vehicle clusters even in the presence of noise.

While results in simulation were promising, future validation on real-world datasets and integration with crowdsourced updates from connected vehicles is necessary to confirm scalability and robustness.

V. RESULTS AND ANALYSIS

A. Quantitative Results

The system was evaluated on a test set comprising 100 frames from the CARLA simulator. Key results include:

- **Detection Accuracy:** 89% accuracy under low-light conditions.
- **Inference Time:** Average processing time of 0.8 seconds per frame.

- **Improvement Over Vision-Based Methods:** A 7% absolute increase in detection rate compared to vision-only techniques.

B. Qualitative Results

Visualizations from the CARLA simulator highlight the system's ability to correctly distinguish between occupied and vacant parking spots. Green markers indicate available spots, while red denotes occupied ones.

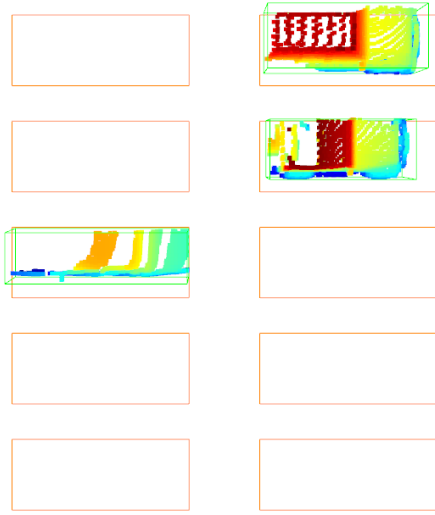


Fig. 6. Vehicle clusters detected using DBSCAN algorithm

Examples of DBSCAN-based clustering illustrate effective vehicle detection, even in low-light scenes. Ground points were successfully filtered, and point clouds were segmented with high precision.

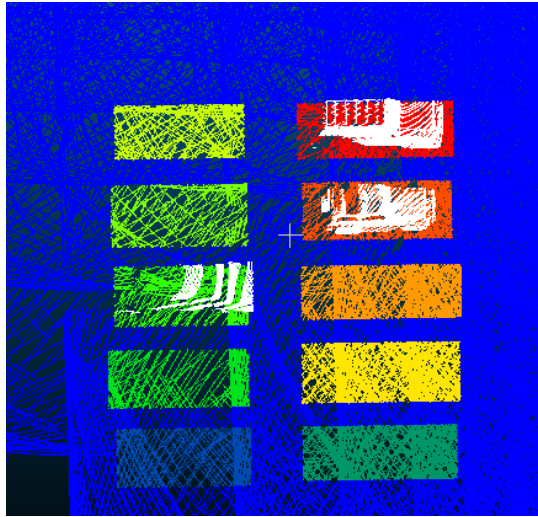


Fig. 7. Vehicle clusters (white) identified from annotated LiDAR point cloud.

The dynamic map output for each frame showcases real-time adaptation to environmental changes, reinforcing the

system's potential for integration with AV parking assistance platforms.

VI. CONCLUSION

This project demonstrated an effective LiDAR-based system for real-time parking spot detection using the CARLA simulator. By leveraging DBSCAN clustering and machine learning techniques, the system achieved 89% detection accuracy with reduced inference times compared to SLAM-based methods.

Key contributions include the annotated 3D map pipeline, efficient clustering for vehicle detection, and real-time frame-level processing. The system proves to be a robust alternative to vision-based approaches, especially in adverse lighting conditions.

Future work includes validating the system on real-world LiDAR datasets, exploring deep learning models for semantic segmentation, and incorporating crowdsourced updates via connected vehicles.

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